

Advanced Topics in Computer Science I

Assignment 2B – Research – Group Topic

Date assigned: Thursday October 2nd, 2025

Date due: Research Notes on individual articles - Monday October 6th, 2025 @10AM

Group Research Notes on 3rd article – Monday October 6th, 2025 @noon

Date due: Group Presentation & Slides - Friday October 10th, 2025 @9AM

No late submissions accepted.

Learning Objectives

Upon successful completion of this assignment, the student will be able to:

- Synthesize relevant information from technical articles,
- Discuss technical topics in a team, and
- Present a technical topic in a team.

Teams:

The teams will be provided by the professor at the beginning of class on Monday October 6th, 2025.

Research Project

In preparation for your individual research presentation, the purpose of this assignment is to read technical articles on a topic, discuss them in a group, synthesize the information and present the most relevant information to the class.

1. This project is to be completed in the groups assigned by the professor. All topics are on Quantum Computing. The groups will be provided in class on Monday October 6th. All members of the group must read three articles – the first two are read by the entire class, and the third article will be chosen by the group.

Read the articles and while reading, make notes of the key points in each article. Submit these notes to Moodle before the beginning of class on Monday October 6th, 2025 in a file called **yourUserId_G40_A02B_Research_Notes**.

Everyone must read both of these articles:

- (1) [What to do while pursuing Quantum Computing](#)
- (2) [Google Quantum Supremacy](#)

2. At the start of class on Monday October 6th, 2025, the groups will be assigned, and you will read the short group specific article. Then you will get together in your groups and discuss the topic and prepare your group notes.

Choose one of these articles to read, in agreement with your group:

- (3) [Exploring Quantum Computing Use Cases for Health Care](#)
- (4) [The Healthy Side of Quantum Computing](#)
- (5) [NASA Quantum Computer Mission](#)
- (6) [Quantum Computing and Cryptography](#)
- (7) [Mercedes Benz Bets on Quantum to Craft the Future of Electric Vehicles](#)
- (8) [Researchers Test Undersea UK-Ireland Quantum Communications Link](#)
- (9) [Toshiba Outlines How Its Building A Quantum-Secure Future](#)

3. By the end of class (noon) on Monday October 6th, 2025, you must submit your group notes on the group article/discussion. This should highlight the key points in the article and the focus of your presentation.
4. Following the discussion, prepare a PowerPoint presentation with your group on your topic. The presentation should be 7 – 9 minutes long and each group member should speak approximately equally. The slides should be visually appealing and contain enough detail to guide you but not be large dumps of text. Include information from the 2 individual articles to help support your group topic. You may use other sources if you wish. If so, they must also be cited.
5. Practice your presentation with your team. The presentations will start at 9:00 AM on Friday October 10th, 2025. The order of the presentations will be provided at the start of class that day.

To submit

- 1) On Monday October 6th, 2025 @10AM submit your notes for the first two individual articles. (No late submissions accepted).
- 2) On Monday October 6th, 2025 @noon submit your group notes for the article your group chose and discussed. (1 submission per group, no late submissions accepted).
- 3) On Friday October 10th, 2025 (before the beginning of class at 9AM), submit your presentation slides (1 submission per group, no late submissions accepted).

Marking Scheme

	Out Of
Research Notes – key points captured on articles in note/bullet form.	15
Group Discussion – clearly demonstrates knowledge of articles, participates in discussion by voicing opinions and asking questions.	5
Group Notes – key points captured on article in note/bullet form.	5
Research Presentation	
Introduction summarizes article and overview.	3
Content – appropriate extraction of key points.	20
Conclusion – main points are restated.	3
Works Cited according to MLA format.	2
Slide format – visually appealing and engaging	7
Delivery of presentation: voice, poise, confidence, eye-contact, professionalism	15
Total	75